

AIM:

Upload and analyse the given dataset and perform data preprocessing techniques and feature scaling.

PYTHON CODE:

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler

data = {
    'Country': ['France', 'Spain', 'Germany', 'Spain', 'Germany',
               'France', 'Spain', 'France', None, 'France'],
    'Age': [44.0, 27.0, 30.0, 38.0, 40.0,
            35.0, None, 48.0, 50.0, 37.0],
    'Salary': [72000.0, 48000.0, 54000.0, 61000.0, None,
               58000.0, 52000.0, 79000.0, 83000.0, 67000.0],
    'Purchased': ['No', 'Yes', 'No', 'No', 'Yes',
                  'Yes', 'No', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

df['Age'] = df['Age'].fillna(df['Age'].mean())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())

X = df[['Age', 'Salary']]

standard_scaler = StandardScaler()
X_standardized = standard_scaler.fit_transform(X)

standardized_df = pd.DataFrame(
    X_standardized,
    columns=['Age', 'Salary']
)

print("\nStandardized Data:")
print(standardized_df)

minmax_scaler = MinMaxScaler()
X_normalized = minmax_scaler.fit_transform(X)

normalized_df = pd.DataFrame(
    X_normalized,
    columns=['Age', 'Salary']
)

print("\nNormalized Data:")
print(normalized_df)
```

OUTPUT:

Standardized Data:

	Age	Salary
0	0.758874	7.494733e-01
1	-1.711504	-1.438178e+00
2	-1.275555	-8.912655e-01
3	-0.113024	-2.532004e-01
4	0.177609	6.632192e-16
5	-0.548973	-5.266569e-01
6	0.000000	-1.073570e+00
7	1.340140	1.387538e+00
8	1.630773	1.752147e+00
9	-0.258340	2.937125e-01

Normalized Data:

	Age	Salary
0	0.739130	0.685714
1	0.000000	0.000000
2	0.130435	0.171429
3	0.478261	0.371429
4	0.565217	0.450794
5	0.347826	0.285714
6	0.512077	0.114286
7	0.913043	0.885714
8	1.000000	1.000000
9	0.434783	0.542857

RESULT:

The given dataset is analysed and preprocessed successfully.